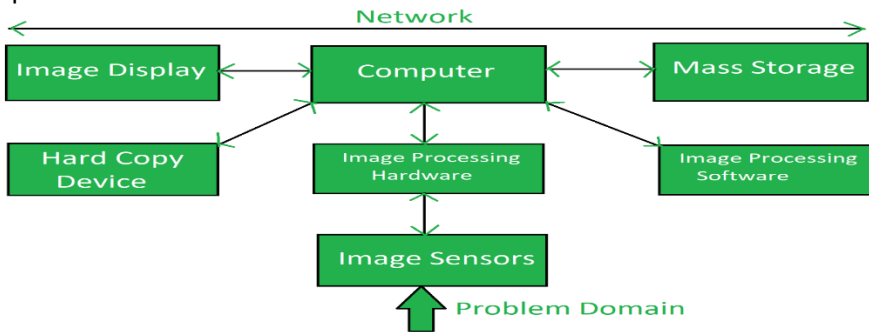


**VII Semester MIDTERM TEST
Computer Vision (CSE_3 0 0 3)****Time Duration: 2 Hours****Date:****Max marks: 30 MARKS**

Question No	Topic	Marks	BL	CO
MCQ				
1	Image processing hardware in the system is mainly used to: a) Provide storage for large image datasets b) Perform parallel arithmetic and logical operations on images c) Convert analog signals into digital signals d) Display processed images	1	Understand	CO1
2	An image of size 1024×1024 pixels is quantized to 16 gray levels. How many bits are required to store the image? a) 1024×1024×4 b) 1024×1024×8 c) 1024×1024×16 d) 1024×1024×2	1	Understand	CO1
3	Compression in image processing is mainly useful for: a) Enhancing contrast b) Storing and transmitting images efficiently c) Reducing segmentation errors d) Detecting shapes in images	1	Understand	CO1
4	If an image is transformed using the negative transformation $s=L-1-r$ which of the following applications benefits most? a) Enhancing details in dark regions of medical X-rays b) Compressing dynamic range of astronomical images c) Equalizing histogram of natural photographs d) Improving efficiency of JPEG compression	1	Understand	CO1
5	Which statement best explains the practical difference between RGB and HSV models? a) HSV is device-dependent, while RGB is perceptually uniform. b) RGB separates chromatic content from intensity, while HSV mixes them. c) HSV is better for segmentation because hue is invariant to illumination changes, unlike RGB. d) RGB is always superior for histogram equalization tasks.	1	Remember	CO1
6	In digital image processing, a pixel represents: a) Image storage unit b) Smallest element of an image c) Image segmentation method d) Wavelet coefficient	1	Remember	CO1

7	Specialized modules and user-written code for processing images belong to: a) Image Processing Software b) Image Display Device c) Mass Storage d) Hard Copy Device	1	Remember	CO1
8	Which of the following transformations is most appropriate to enhance details in dark regions of a medical image? a) Logarithmic transformation. b) Negative transformation. c) Power-law with $\gamma > 1$. d) Bit-plane slicing of the MSB.	1	Remember	CO1
9	In industrial applications, thermal imaging is primarily used for: a) Crop monitoring b) Detecting material defects c) Identifying emotions d) Enhancing entertainment videos	1	Understand	CO1
10	Which component provides a general-purpose platform for running image processing tasks? a) Image Sensor b) Computer c) Hard Copy Device d) Network	1	Remember	CO1
Descriptive Questions				
11a	Analyze the role of Power-Law Transformation and Logarithmic Transformation in gray-level image enhancement. With the help of suitable graphs, discuss how each transformation modifies the dynamic range of dark and bright regions, and explain why such enhancements are necessary in industrial quality inspection of materials and defect detection Logarithmic Transformation-1 Mark Power-Law Transformation -1 Mark Graphical Explanation- 2 Marks Industrial Application & Justification-1 Mark	5	Analyze	CO1
11b	A hospital uses computer vision to automatically detect tumors in medical scans. Describe how each step of digital image processing is applied in this system In tumor detection, images are first preprocessed for clarity, segmented to isolate abnormal areas, and then analyzed using classifiers to identify and label tumor regions automatically. Explain all the steps-3 Marks	3	Apply	CO1
11c	Apply image negation on the following image of size 4x4 and write the output image matrix. $I = \begin{bmatrix} 0 & 50 & 128 & 255 \\ 30 & 90 & 180 & 220 \\ 10 & 40 & 60 & 200 \\ 15 & 100 & 150 & 240 \end{bmatrix}$ Image negation for an 8-bit grayscale image is computed as $I' = 255 - I$	2	Apply	CO1

	$I' = \begin{bmatrix} 255 & 205 & 127 & 0 \\ 225 & 165 & 75 & 35 \\ 245 & 215 & 195 & 55 \\ 240 & 155 & 105 & 15 \end{bmatrix}$			
12a	Consider the following image matrix. Compute the smallest 4-,8 and m-path in the given image matrix and also mention the path length. Consider $V=\{2,3,5\}$ Smallest 4-path restricted to pixels V=no 4-path-1 mark Smallest 8-path restricted to pixels V=3→2→3→2→3-1 mark Smallest m-path restricted to pixels V=3 →2 → 3 →2 →3.-1 mark Write all the steps in detailed way and mark the paths in proper way—2 marks 	5	Apply	CO1
12b	An automobile company uses an automated vision system to inspect defects in car parts. The system captures part images, processes them to detect scratches or cracks, stores inspection results, and generates quality reports. Identify and explain the role of each component in the image processing system in this scenario. Diagram-1 Marks Explanation-2 Marks 	3	Apply	CO1
12c	Discuss how pixel intensity values are mapped to discrete levels and its effect on image quality. In digital images, continuous intensity values from sensors are quantized into discrete gray levels . Each pixel's brightness is represented by a numerical value between 0 (black) and L-1 (white). Higher number of gray levels → smoother tones and better detail. Lower levels → loss of detail due to insufficient intensity representation.	2	Create	CO1

Note: BL refers to Bloom's Taxonomy Level.